MSG No. 111.252 $P\bar{4}2m1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-42m1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `1b`, site symmetry: `-42m1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `1c`, site symmetry: `-42m1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 4: Wyckoff site: `1d`, site symmetry: `-42m1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]`

Table 5: Wyckoff site: `2e`, site symmetry: `222.1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[5,6,7,8,13,14,15,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `222.1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 7: Wyckoff site: `2g`, site symmetry: `2.mm1'`

No.	position	mapping
1	$[0, 0, z]$	`[1,4,7,8,9,12,15,16]`
2	$[0, 0, -z]$	`[2,3,5,6,10,11,13,14]`

Table 8: Wyckoff site: `2h`, site symmetry: `2.mm1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[1,4,7,8,9,12,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[2,3,5,6,10,11,13,14]`

Table 9: Wyckoff site: `4i`, site symmetry: `.2.1'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2,9,10]`
2	$[-x, 0, 0]$	`[3,4,11,12]`
3	$[0, -x, 0]$	`[5,7,13,15]`
4	$[0, x, 0]$	`[6,8,14,16]`

Table 10: Wyckoff site: `4j`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,9,10]`
2	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[3,4,11,12]`
3	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	`[5,7,13,15]`
4	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	`[6,8,14,16]`

Table 11: Wyckoff site: `4k`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,2,9,10]`
2	$\left[-x, 0, \frac{1}{2}\right]$	`[3,4,11,12]`
3	$\left[0, -x, \frac{1}{2}\right]$	`[5,7,13,15]`
4	$\left[0, x, \frac{1}{2}\right]$	`[6,8,14,16]`

Table 12: Wyckoff site: `4l`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	`[1,2,9,10]`
2	$\left[-x, \frac{1}{2}, 0\right]$	`[3,4,11,12]`
3	$\left[\frac{1}{2}, -x, 0\right]$	`[5,7,13,15]`
4	$\left[\frac{1}{2}, x, 0\right]$	`[6,8,14,16]`

Table 13: Wyckoff site: `4m`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,4,9,12]`
2	$\left[0, \frac{1}{2}, -z\right]$	`[2,3,10,11]`
3	$\left[\frac{1}{2}, 0, -z\right]$	`[5,6,13,14]`

continued ...

Table 13

No.	position	mapping
4	$\left[\frac{1}{2}, 0, z\right]$	[7,8,15,16]

Table 14: Wyckoff site: 4n, site symmetry: ..m1'

No.	position	mapping
1	$[x, x, z]$	[1,8,9,16]
2	$[x, -x, -z]$	[2,5,10,13]
3	$[-x, x, -z]$	[3,6,11,14]
4	$[-x, -x, z]$	[4,7,12,15]

Table 15: Wyckoff site: 8o, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,9]
2	$[x, -y, -z]$	[2,10]
3	$[-x, y, -z]$	[3,11]
4	$[-x, -y, z]$	[4,12]
5	$[y, -x, -z]$	[5,13]
6	$[-y, x, -z]$	[6,14]
7	$[-y, -x, z]$	[7,15]
8	$[y, x, z]$	[8,16]